Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details

Batch Number:13

Start date: 02-Feb-26

Names:

S.no	Name
1	Romen Halder
2	Balakuberan KS
3	Aaradhya Battini

Internship Duration: 8 Weeks

Project Title

PharmaTrack Pro – Intelligent Pharmacy Inventory & Compliance Management System

1. Project Objective

Design and develop a centralized pharmacy inventory management system to digitally manage medicines, stock records, and pharmacy transactions.

Enable real-time monitoring of medicine inventory including batch numbers, expiry dates, and stock levels.

Reduce medication errors and stock discrepancies by replacing manual pharmacy record systems with an automated digital solution.

Implement role-based authentication and authorization to ensure secure access for administrators, managers, and pharmacists.

Develop a transaction management module to record medicine sales, purchase entries, and stock updates with complete traceability.

Ensure regulatory compliance by maintaining digital prescription records and logs for controlled medicines such as Schedule H and H1 drugs.

2. Project description in detail

PharmaTrack Pro – Intelligent Pharmacy Inventory & Compliance Management System is a modern digital platform designed to simplify and automate pharmacy management operations. Many pharmacies still rely on manual registers or basic spreadsheet systems for managing medicine inventory, prescriptions, and sales records. These outdated methods often lead to medication errors, stock mismanagement, compliance risks, and lack of real-time insights.

PharmaTrack Pro addresses these challenges by offering a centralized and automated pharmacy management solution that ensures accurate tracking of medicines, improved regulatory compliance, and enhanced operational efficiency.

The system enables authorized users to manage medicines and pharmacy stock through a structured interface. Each medicine entry contains detailed attributes including medicine name, category, batch number, expiry date, quantity, supplier information, and pricing details. Pharmacists can easily add new medicines, update stock levels, remove expired products, and search medicines instantly.

One of the core features of PharmaTrack Pro is the **FEFO (First Expiry, First Out) logic**, which ensures that medicines nearing expiry are automatically prioritized for sale. This helps pharmacies minimize financial loss caused by expired medications.

The platform also includes a **secure authentication system with role-based access control**. Administrators can manage employees and system configurations, while pharmacists can handle medicine sales and inventory operations. This ensures controlled access and accountability within the system.

Another important feature is the **transaction management system**, which records every sale and inventory adjustment with detailed logs. Each transaction includes medicine details, quantity sold, batch number, timestamp, and user information. The system automatically updates inventory levels after each transaction to maintain accurate stock records.

PharmaTrack Pro also includes **regulatory compliance features**. For controlled drugs such as Schedule H and H1 medicines, the system requires prescription verification and doctor registration details before allowing the sale. This helps pharmacies maintain compliance with drug control regulations.

To assist pharmacy owners in decision-making, the system provides a **smart dashboard with analytics and visualizations**. The dashboard displays key insights such as total medicines in stock, low-stock alerts, medicines nearing expiry, category-wise distribution, and sales trends.

Additionally, an **inventory alert system** notifies pharmacists and managers when stock levels fall below predefined limits or when medicines are close to expiry. This ensures timely restocking and prevents stock shortages.
The system is designed with a scalable architecture and can be used by **retail pharmacies, hospital pharmacies, medical stores, and pharmaceutical distributors**.

3. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Requirement analysis, problem identification, and system architecture design.	Studied pharmacy management challenges, finalized system requirements, and designed architecture and module structure
Week 2	Database schema design and backend setup	Designed relational database schema for medicines, users, batches, and transactions; configured backend environment
Week 3	Authentication system and role-based access control	Implemented secure login system, password encryption, and role-based access for Admin, Manager, and Pharmacist
Week 4	Medicine and inventory management module	Developed medicine addition, batch tracking, expiry management, and stock update features
Week 5	Transaction and sales module	Implemented POS system for medicine sales with automatic stock deduction and transaction logging
Week 6	Inventory alert system and notification module	Developed low-stock alerts and expiry notifications for pharmacy staff
Week 7	Dashboard and analytics development	Built dashboard displaying stock statistics, expiry insights, and medicine category analysis
Week 8	System testing, debugging, documentation, and deployment	Conducted testing, fixed bugs, optimized performance, completed documentation, and demonstrated the system

4. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Initial project briefing, pharmacy workflow analysis, and technology stack selection	Week-1

Milestone	Description	Date Achieved
Database & Backend Setup Completed	Database schema creation and implementation of backend APIs	Week-2
Authentication Module Completed	Secure login system with JWT authentication and role-based access	Week-3
Inventory Module Prototype Ready	Implementation of medicine inventory management with batch tracking	Week-4
Transaction Tracking System Completed	Implementation of low-stock alerts and expiry notifications	Week-5
Dashboard & Analytics Integration	Interactive dashboard with inventory insights and graphical charts	Week-6
System Testing & Optimization Completed	Completion of functional testing, bug fixes, and performance improvements	Week-7
Final System Deployment & Documentation	Integration of all modules, preparation of documentation, and final project demonstration	Week-8

5. Project execution details

The **PharmaTrack Pro** system was developed using a modern full-stack architecture to ensure performance, scalability, and security. The system was divided into key modules including authentication, medicine inventory management, transaction management, alert notifications, and analytics dashboard.

The backend was implemented using **Java with Spring Boot**, which provides a robust framework for developing RESTful APIs. These APIs manage business logic, process user requests, and interact with the database. The database used for the system is **PostgreSQL (via Supabase)**, which stores information related to users, medicines, batch details, transactions, and compliance records.

The database schema was designed using normalized relational tables with appropriate foreign key relationships to maintain data integrity. Key tables include medicines, batches, users, doctors, and transactions.

A secure authentication system was implemented using **Spring Security and JWT (JSON Web Token)**. User credentials are securely stored using **Bcrypt password hashing**. Role-based access control ensures that different user roles such as Admin, Manager, and Pharmacist have appropriate permissions within the system. The **inventory management module** allows users to add medicines, update stock quantities, track batch numbers, and monitor expiry dates. Backend validations ensure accurate stock updates and prevent invalid data entries.

The **transaction module** handles medicine sales and stock updates. Each sale records medicine details, batch number, quantity sold, user ID, and timestamp. The system automatically deducts sold quantities from inventory and updates stock levels in real time.

PharmaTrack Pro also includes an **inventory alert system** that monitors stock levels and expiry dates. When stock levels fall below the defined threshold or medicines are nearing expiry, alerts are generated to notify pharmacy staff.

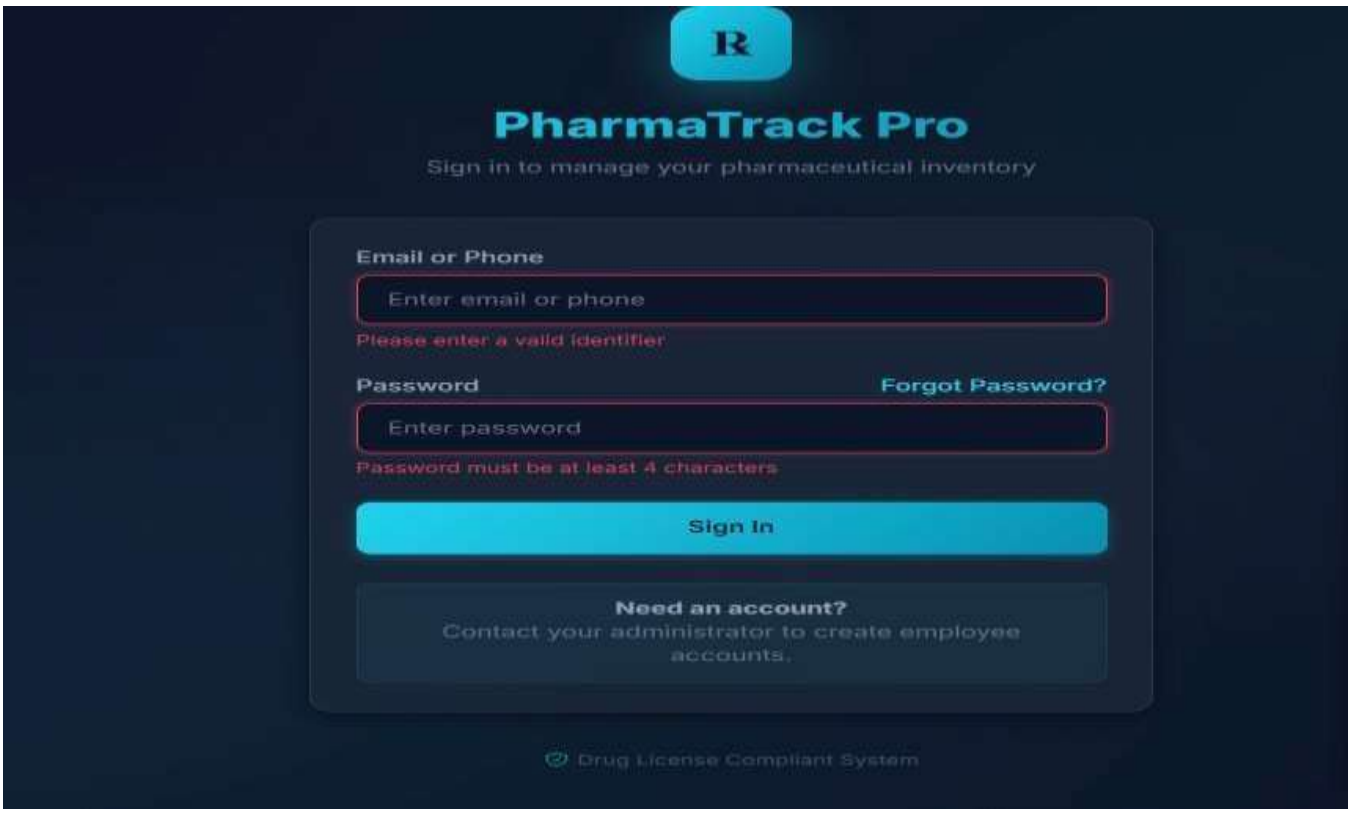
The frontend was developed using **React.js**, providing a responsive and user-friendly interface for pharmacists and administrators. The interface includes modules for login, inventory management, transaction history, and analytics dashboards.

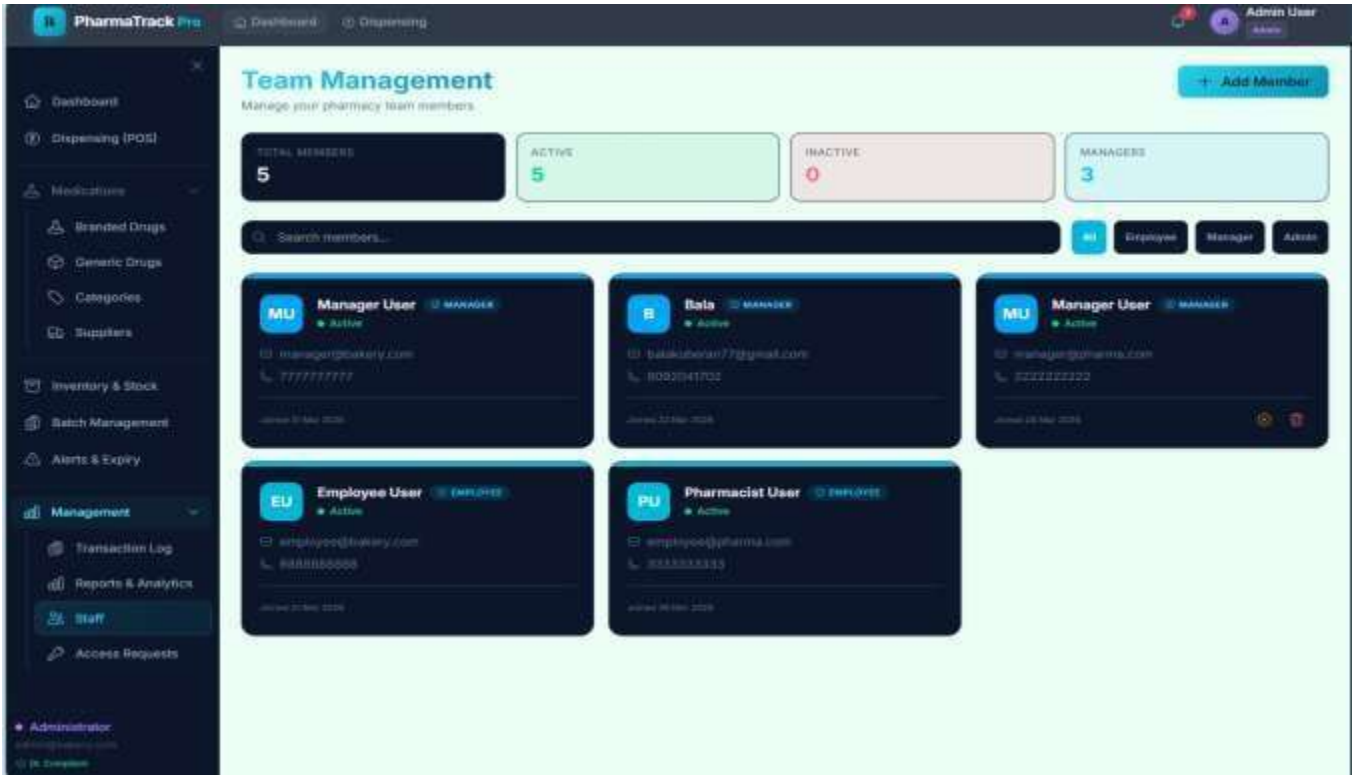
The dashboard uses visual charts and summary cards to present important information such as stock health, low-stock medicines, expiry alerts, and category-based distribution.

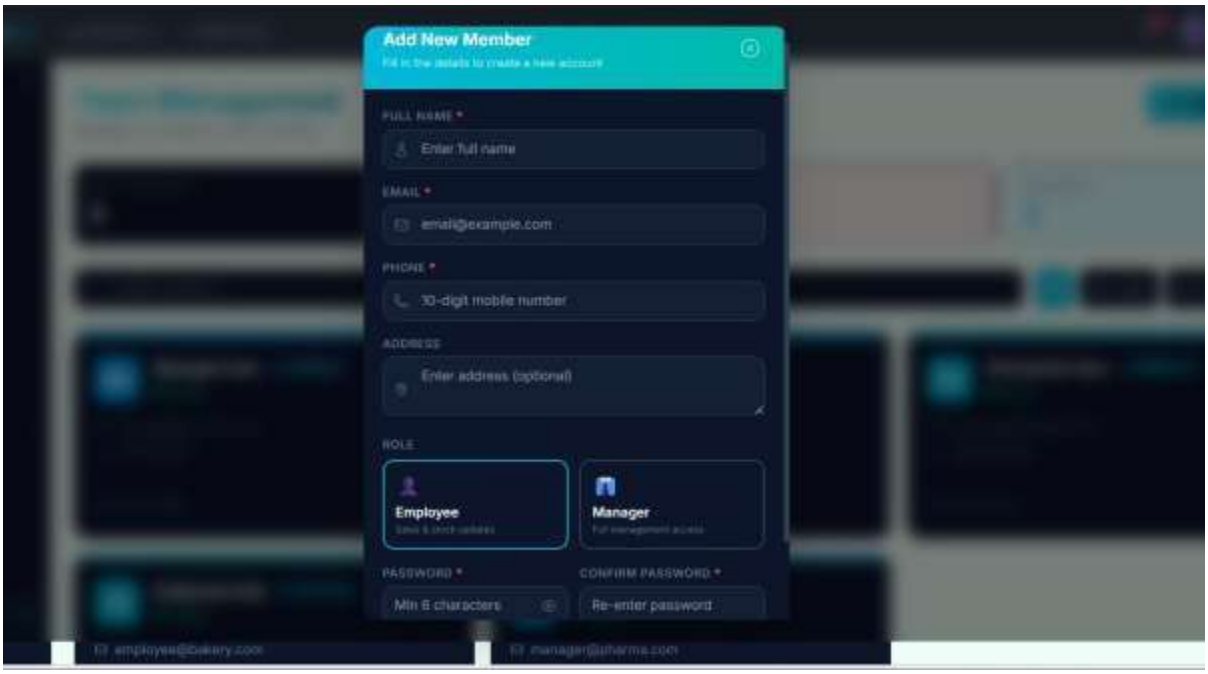
Testing was conducted at multiple stages including **backend API testing, frontend interface testing, and system integration testing**. Performance improvements were achieved through optimized database queries, indexed columns, and efficient API response handling.

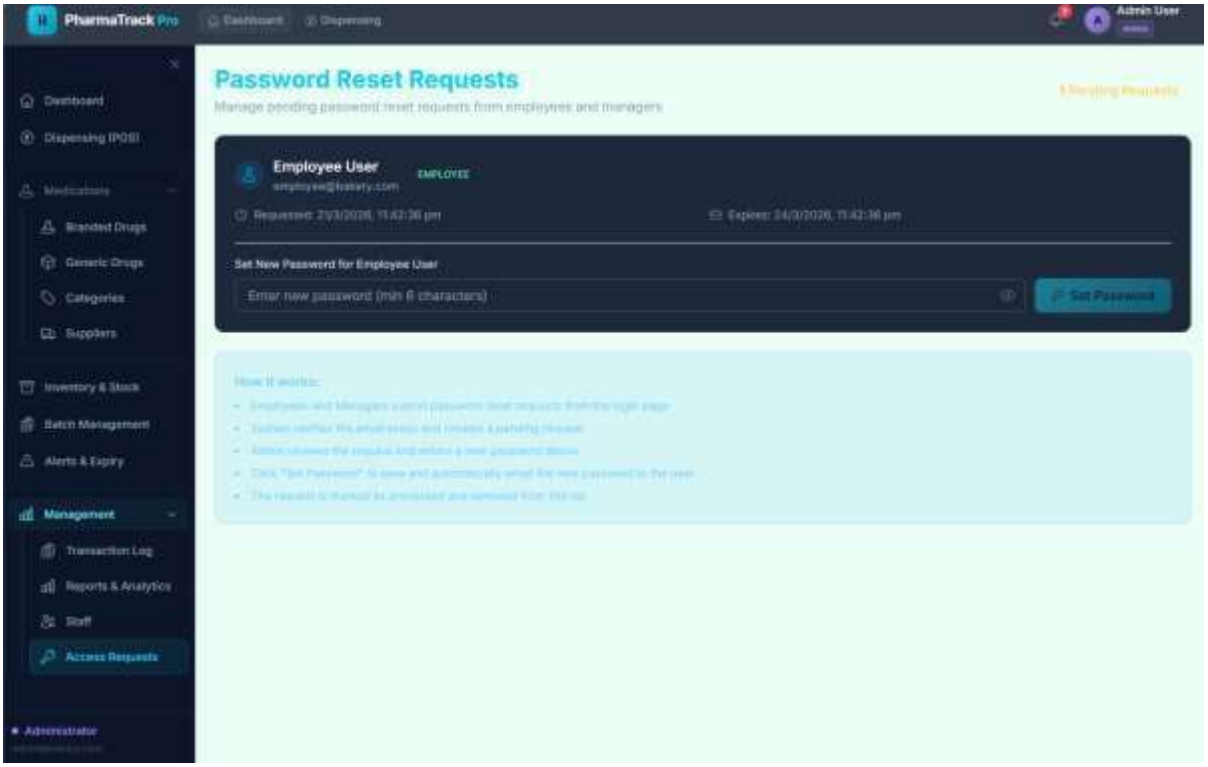
Finally, all modules were integrated and deployed in a test environment. Comprehensive **technical documentation and a user guide** were prepared to ensure easy system usage and future scalability.

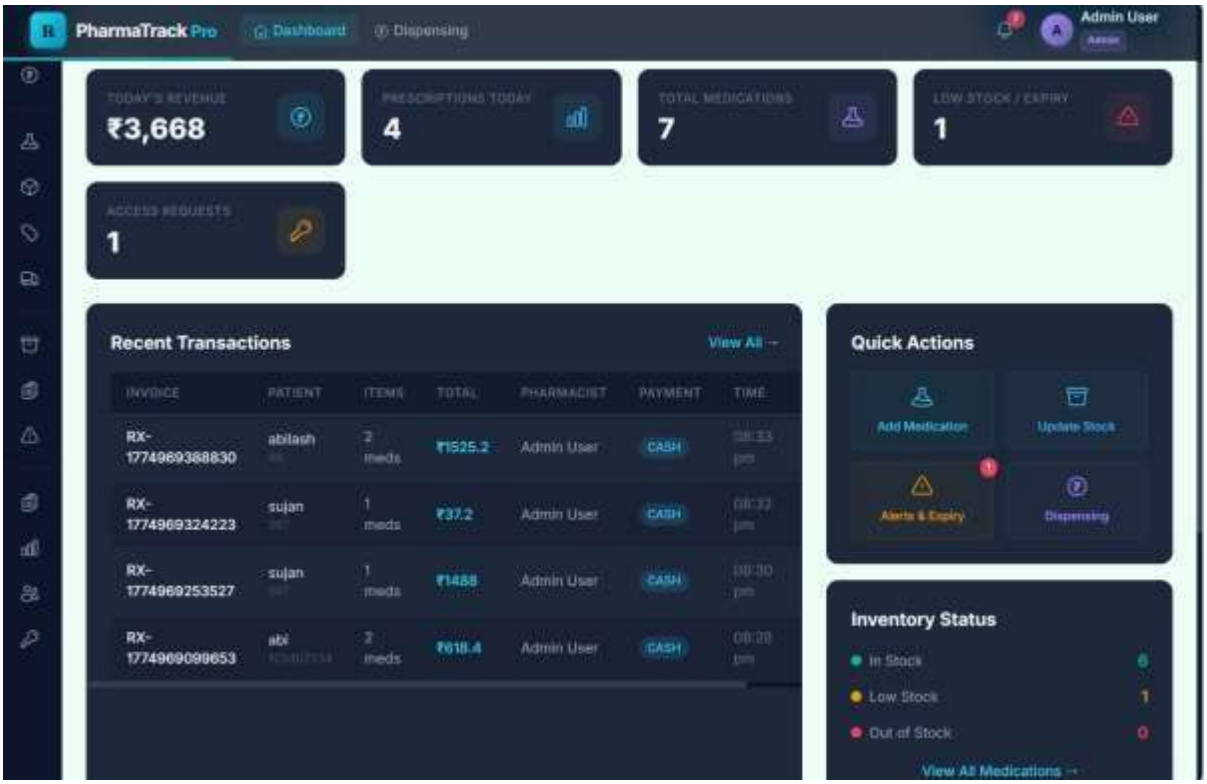
6. Snapshots / Screenshots

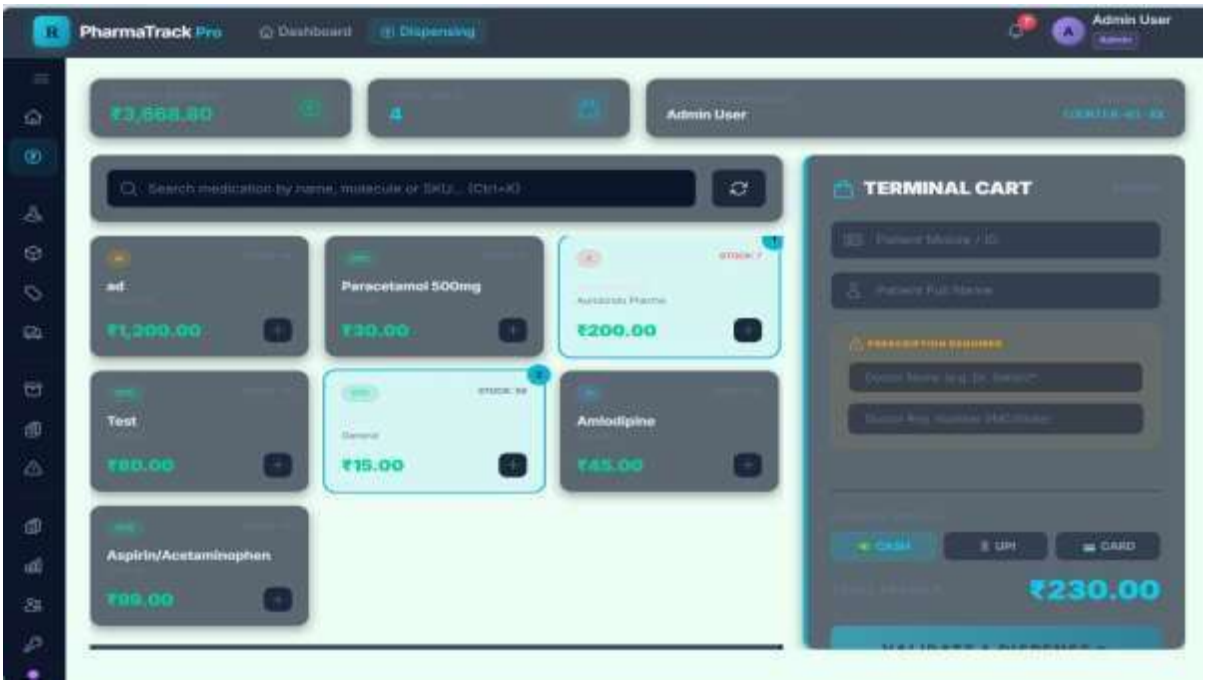


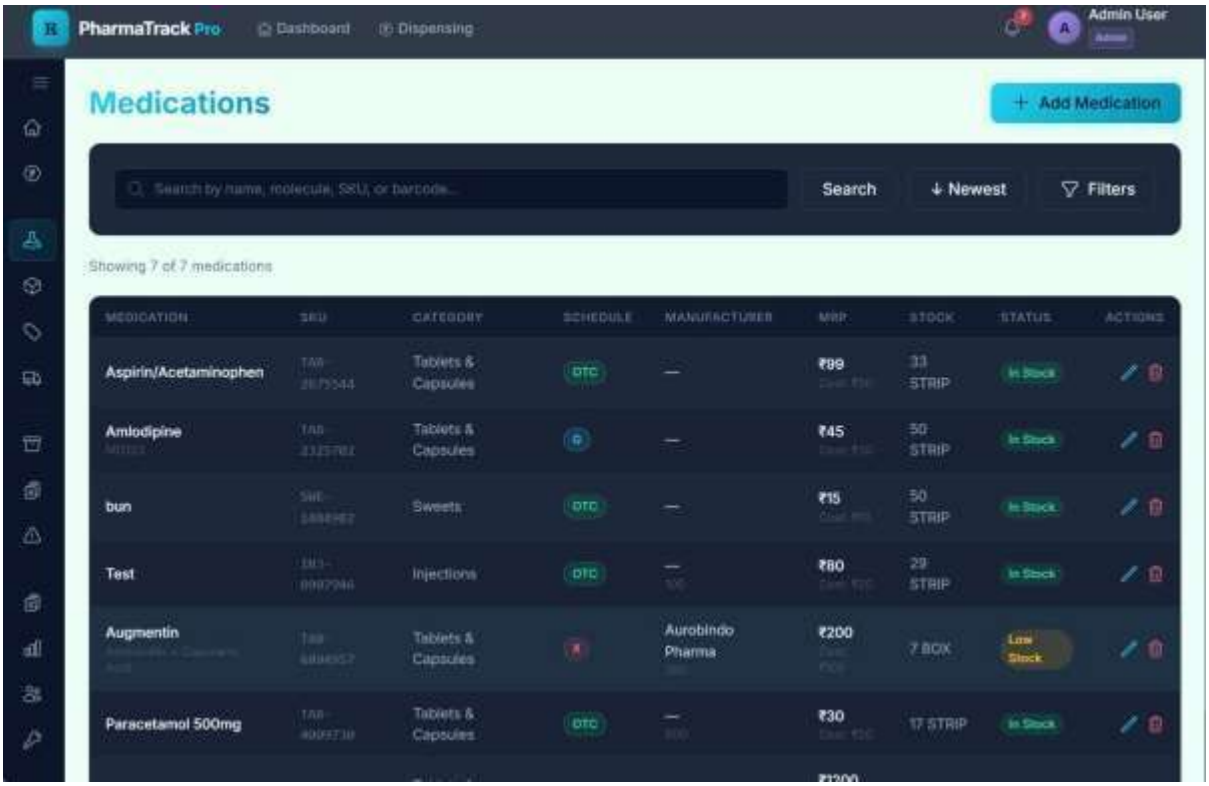


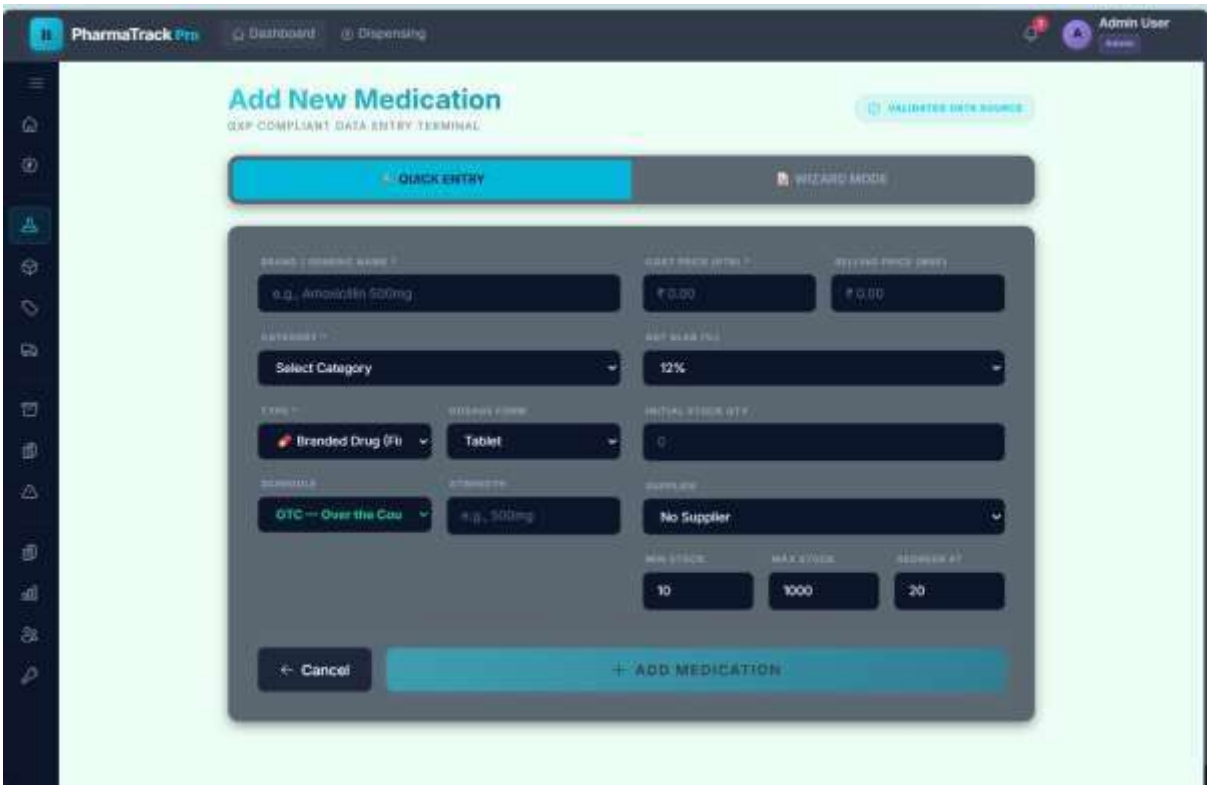

Add New Member
Fill in the details to create a new account
FULL NAME *
Enter full name
EMAIL *
email@example.com
PHONE *
10-digit mobile number
ADDRESS
Enter address (optional)
ROLE
Employee
Manager
PASSWORD *
Min 6 characters
CONFIRM PASSWORD *
Re-enter password

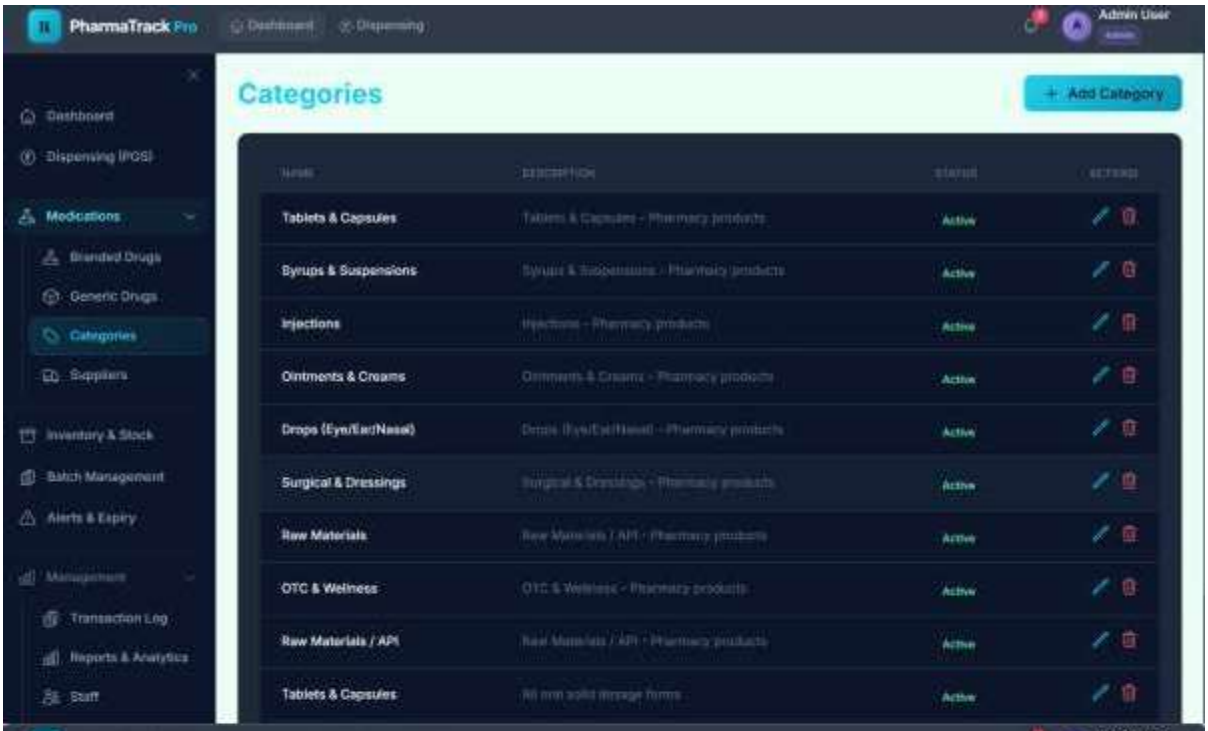

PharmaTrack Pro
Dashboard
Dispensing
Admin User
Dashboard
Dispensing (POS)
Medications
Branded Drugs
Generic Drugs
Categories
Suppliers
Inventory & Stock
Batch Management
Alerts & Expiry
Management
Transaction Log
Reports & Analytics
Staff
Access Requests
Administrator
Password Reset Requests
Manage pending password reset requests from employees and managers
Employee User
EMPLOYEE
Set New Password for Employee User
Enter new password (min 6 characters)
Set Password

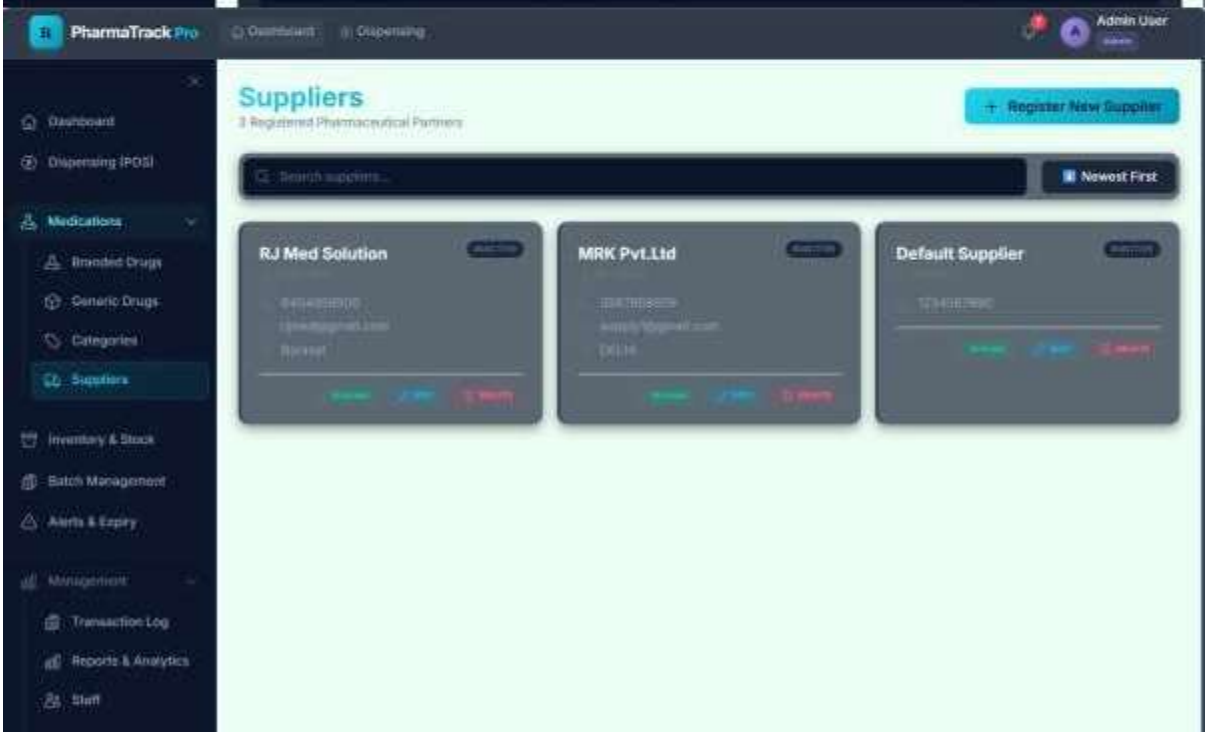

PharmaTrack Pro
Dashboard
Dispensing
Admin User
TODAY'S REVENUE
₹3,668
PRESCRIPTIONS TODAY
4
TOTAL MEDICATIONS
7
LOW STOCK / EXPIRY
1
ACCESS REQUESTS
1
Recent Transactions
View All →
INVOICE
PATIENT
ITEMS
TOTAL
PHARMACIST
PAYMENT
TIME
RX-1774969388830
abilash
2 meds
₹1525.2
Admin User
CASH
RX-1774969324223
sujan
1 meds
₹37.2
Admin User
CASH
RX-1774969253527
sujan
1 meds
₹1488
Admin User
CASH
RX-1774969099653
abi
2 meds
₹618.4
Admin User
CASH
Quick Actions
Add Medication
Update Stock
Alerts & Expiry
Dispensing
Inventory Status
In Stock
Low Stock
1
Out of Stock
0
View All Medications →


PharmaTrack Pro
Dashboard
Dispensing
Admin User
4
Admin User
TERMINAL CART
ad
₹1,200.00
Paracetamol 500mg
₹30.00
₹200.00
Test
₹15.00
Amlodipine
₹65.00
Aspirin/Acetaminophen
₹99.00
CASH
UPI
CARD
₹230.00


PharmaTrack Pro
Dashboard
Dispensing
Admin User
Medications
+ Add Medication
Search by name, molecule, SKU, or barcode...
Search
↓ Newest
Filters
Showing 7 of 7 medications
MEDICATION
SKU
CATEGORY
SCHEDULE
MANUFACTURER
MRP
STOCK
STATUS
ACTIONS
Aspirin/Acetaminophen
Tablets & Capsules
OTC
₹99
33 STRIP
In Stock
Amlodipine
Tablets & Capsules
₹45
50 STRIP
In Stock
bun
Sweets
OTC
₹15
50 STRIP
In Stock
Test
Injections
OTC
₹80
29 STRIP
In Stock
Augmentin
Tablets & Capsules
Aurobindo Pharma
₹200
7 BOX
Low Stock
Paracetamol 500mg
Tablets & Capsules
OTC
₹30
17 STRIP
In Stock


PharmaTrack Pro
Dashboard
Dispensing
Admin User
Add New Medication
GXP COMPLIANT DATA ENTRY TERMINAL
QUICK ENTRY
WIZARD MODE
e.g., Amoxicillin 500mg
₹ 0.00
₹ 0.00
Select Category
12%
Branded Drug (Fi
Tablet
0
OTC — Over the Cou
e.g., 500mg
No Supplier
10
1000
20
← Cancel
+ ADD MEDICATION


PharmaTrack Pro
Dashboard
Dispensing
Admin User
Dashboard
Dispensing (POS)
Medications
Branded Drugs
Generic Drugs
Categories
Suppliers
Inventory & Stock
Batch Management
Alerts & Expiry
Management
Transaction Log
Reports & Analytics
Staff
Categories
+ Add Category
Tablets & Capsules
Active
Syrups & Suspensions
Active
Injections
Active
Ointments & Creams
Active
Drops (Eye/Ear/Nasal)
Active
Surgical & Dressings
Active
Raw Materials
Active
OTC & Wellness
Active
Raw Materials / API
Active
Tablets & Capsules
Active


PharmaTrack Pro
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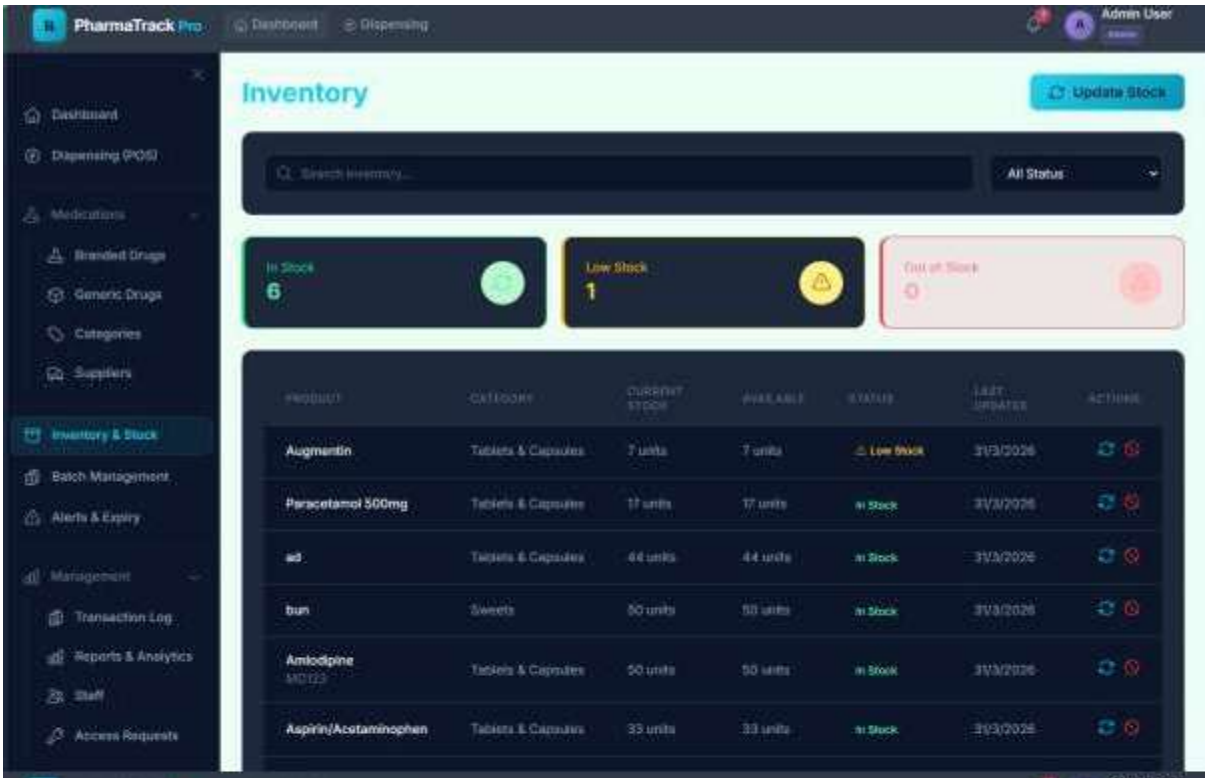
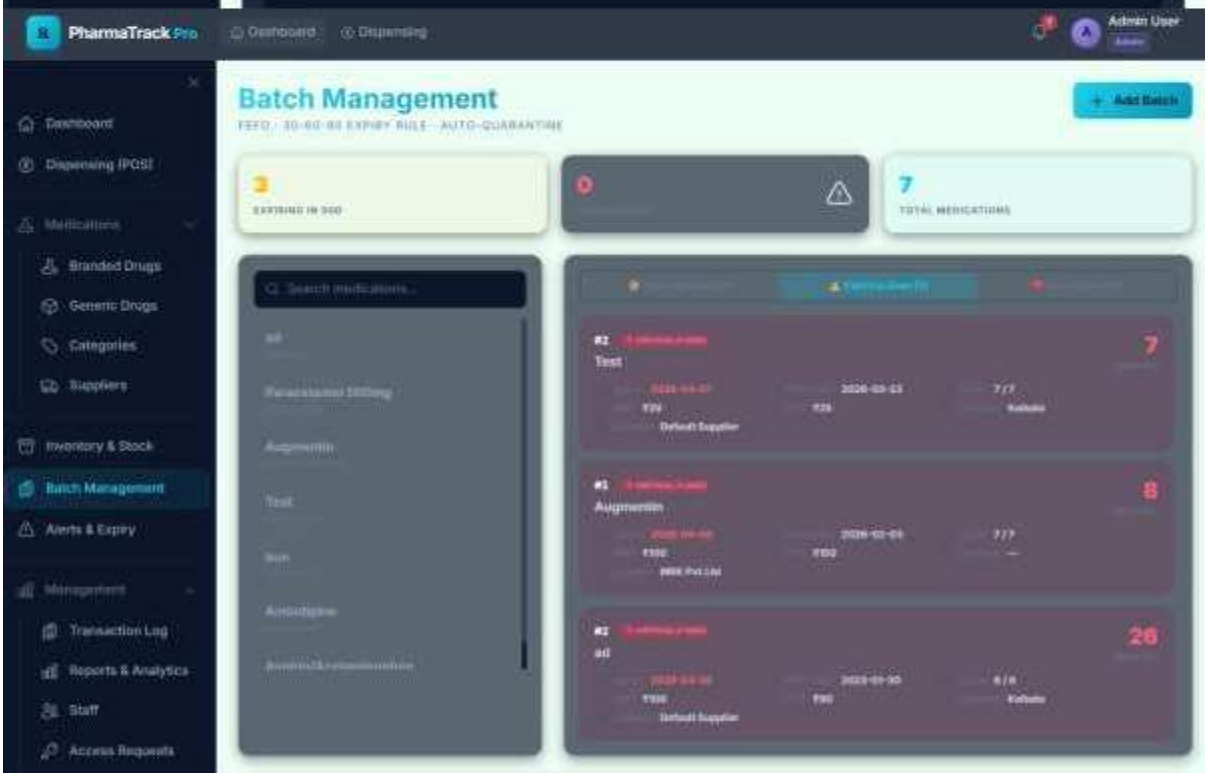
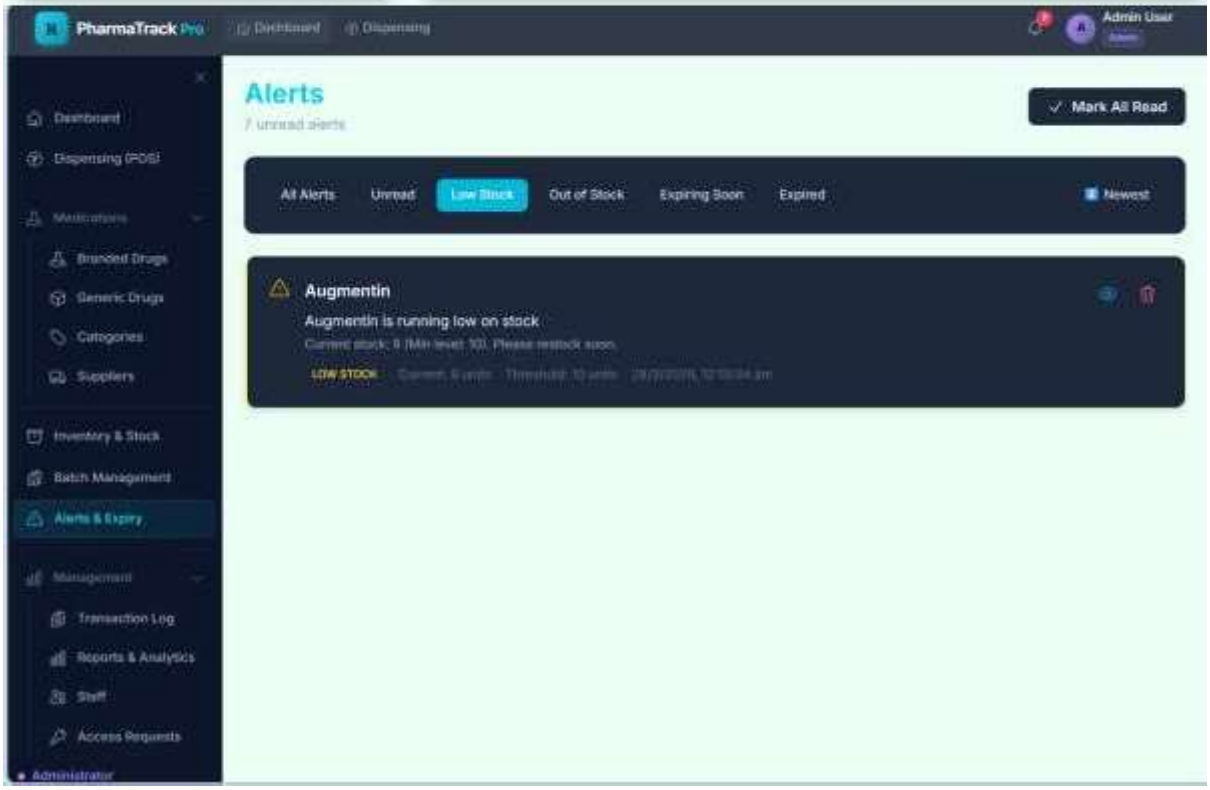
Medications
Branded Drugs
Generic Drugs
Categories
Suppliers
Inventory & Stock
Batch Management
Alerts & Expiry
Management
Transaction Log
Reports & Analytics
Staff
Suppliers
3 Registered Pharmaceutical Partners
+ Register New Supplier
Search suppliers...
Newest First
RJ Med Solution
MRK Pvt.Ltd
Default Supplier


PharmaTrack Pro
Dashboard
Dispensing
Admin User
Inventory
Update Stock
All Status
In Stock
6
Low Stock
1
Out of Stock
0
Dashboard
Dispensing (POS)
Medications
Branded Drugs
Generic Drugs
Categories
Suppliers
Inventory & Stock
Batch Management
Alerts & Expiry
Management
Transaction Log
Reports & Analytics
Staff
Access Requests
PRODUCT
CATEGORY
CURRENT STOCK
AVAILABLE
STATUS
LAST UPDATED
ACTIONS
Augmentin
Tablets & Capsules
7 units
7 units
Low Stock
31/3/2026
Paracetamol 500mg
Tablets & Capsules
17 units
17 units
In Stock
31/3/2026
ad
Tablets & Capsules
44 units
44 units
In Stock
31/3/2026
bun
Sweets
50 units
50 units
In Stock
31/3/2026
Amlodipine
Tablets & Capsules
50 units
50 units
In Stock
31/3/2026
Aspirin/Acetaminophen
Tablets & Capsules
33 units
33 units
In Stock
31/3/2026


PharmaTrack Pro
Dashboard
Dispensing
Admin User
Batch Management
Add Batch
3
0
7
TOTAL MEDICATIONS
Dashboard
Dispensing (POS)
Branded Drugs
Generic Drugs
Categories
Suppliers
Inventory & Stock
Batch Management
Alerts & Expiry
Transaction Log
Reports & Analytics
Staff
Access Requests
Test
7
Augmentin
8
ad
26

Search medications...
Paracetamol 500mg
Augmentin
Test
Amlodipine
#INIT-TAB-4009730-F06FE0
726
₹30
₹30
MRK Pvt.Ltd


PharmaTrack Pro
Dashboard
Dispensing
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Dispensing (POS)
Medications
Branded Drugs
Generic Drugs
Categories
Suppliers
Inventory & Stock
Batch Management
Alerts & Expiry
Management
Transaction Log
Reports & Analytics
Staff
Access Requests
Administrator
Alerts
7 unread alerts
Mark All Read
All Alerts
Unread
Low Stock
Out of Stock
Expiring Soon
Expired
Newest
Augmentin
Augmentin is running low on stock
LOW STOCK

PharmaTrack Pro | Dashboard | Dispensing | Admin User

Dashboard
Dispensing (POS)
Medications
Branded Drugs
Generic Drugs
Categories
Suppliers
Inventory & Stock
Batch Management
Alerts & Expiry
Management
Transaction Log
Reports & Analytics
Staff
Access Requests

Transaction Log

Track all inventory movements

Refresh

Total Stock In	Total Stock Out	Total Wastage	Total Transactions
233	21	20	28

Period: Today | This Week | This Month | All Time | Type: All Types | Search by product, user, reference | Newest First

Showing 28 transactions

DATE & TIME	PRODUCT	TYPE	QTY CHANGE	BEFORE → AFTER	BY	REFERENCE
31 Mar 2026	Test	Wastage	-20	49 → 29	Admin User	-
31 Mar 2026	Test	Stock In	+30	19 → 49	Admin User	-
31 Mar 2026	Aspirin/Acetaminophen	Stock In	+33	0 → 33	Admin User	New batch received: INT-TAB-267
31 Mar 2026	Amlodipine	Stock In	+50	0 → 50	Admin User	New batch received: INT-TAB-232
31 Mar 2026	bun	Stock In	+50	0 → 50	Admin User	New batch received: INT-SWE-144
31 Mar 2026	ad	Stock Out	-1	45 → 44	Admin User	SALE - Order: null

Reports & Analytics

Business insights and downloadable reports

Download PDF

Analytics | Stock Report | Sales Report | Usage Report

Start Date: 01-03-2026
End Date: 31-03-2026
Generate | CSV

Total	Stock In	Stock Out	Adjustments	Wastage
29	10	18	0	1

Transaction Details

28 transactions

DATE	PRODUCT	TYPE	QTY	PREVIOUS	NEW	USER	REASON
31 Mar 2026	Test	WASTAGE	-20	49	29	Admin User	-
31 Mar 2026	Test	STOCK IN	30	19	49	Admin User	-
31 Mar 2026	Aspirin/Acetaminophen	STOCK IN	33	0	33	Admin User	New batch received: I...
31 Mar 2026	Amlodipine	STOCK IN	50	0	50	Admin User	New batch received: I...
31 Mar 2026	bun	STOCK IN	50	0	50	Admin User	New batch received: I...
31 Mar 2026	ad	STOCK OUT	-1	45	44	Admin User	SALE - Order: null
31 Mar 2026	Paracetamol 500mg	STOCK OUT	-1	18	17	Admin User	SALE - Order: null

Reports & Analytics

Business insights and downloadable reports

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Analytics | Stock Report | Sales Report | Usage Report

Start Date: 01-03-2026

End Date: 31-03-2026

Generate | CSV

Total Products	Finished Goods	Raw Materials	Low Stock	Out of Stock
7			1	0

Stock Details

7 items

PRODUCT	CATEGORY	TYPE	STOCK	MIN STOCK	STATUS
Augmentin	Tablets & Capsules	Raw	7	0	LOW STOCK
Paracetamol 500mg	Tablets & Capsules	Raw	17	0	IN STOCK
ad	Tablets & Capsules	Raw	44	0	IN STOCK
bun	Sweets	Raw	50	0	IN STOCK
Amlodipine	Tablets & Capsules	Raw	50	0	IN STOCK
Aspirin/Acetaminophen	Tablets & Capsules	Raw	33	0	IN STOCK
Test	Injections	Raw	29	0	IN STOCK

7. Learnings & Skills Acquired

Gained practical experience in designing and managing relational databases using PostgreSQL for storing pharmacy inventory, batch records, and transaction logs

• Developed strong understanding of backend development using Java and Spring Boot for building secure RESTful APIs

• Learned to implement secure authentication using JWT and Spring Security for protecting system access

• Acquired skills in frontend development using React.js for building responsive and interactive pharmacy management interfaces

• Improved knowledge of API communication and integration between frontend and backend systems

• Understood real-world implementation of pharmacy inventory logic such as batch tracking, expiry monitoring, and FEFO (First Expiry First Out) stock rotation

• Learned to design data dashboards and analytics to visualize stock status, low inventory alerts, and pharmacy sales insights

• Enhanced ability in system debugging, testing, and performance optimization

• Gained experience in full-stack application architecture and module integration

• Improved team collaboration, documentation writing, and project presentation skills

8. Challenges Face

Challenge 1: Designing an Efficient Database Structure

Problem: Managing relationships between users, inventory items, and transactions without data redundancy

Solution: Designed a normalized MySQL schema with proper foreign keys and indexing to maintain data integrity

Challenge 2: Maintaining Accurate Real-Time Stock Updates

Problem: Ensuring inventory quantities remain correct after every issue and return transaction

Solution: Implemented backend logic in Java to automatically update stock levels and validate quantity constraints

Challenge 3: Implementing Secure Role-Based Access Control

Problem: Preventing unauthorized users from accessing or modifying sensitive inventory data

Solution: Developed a role-based authentication system with encrypted passwords and controlled API access

Challenge 4: Integrating React Frontend with Java Backend

Problem: Handling smooth data exchange between frontend and backend services

Solution: Built RESTful APIs and implemented proper error handling and response management for seamless integration

9. Testimonials from team

Working on the **PharmaTrack Pro project** was an enriching experience for the entire team. Each team member contributed to different modules such as backend development, frontend interface design, and database management, which helped us understand how a real-world pharmacy management system is built and integrated.
Through continuous collaboration, planning discussions, and debugging sessions, we were able to overcome technical challenges effectively. The project improved our practical knowledge of **Java, Spring Boot, PostgreSQL, and React**, while also strengthening our ability to solve real-world software problems.
This project also enhanced our **teamwork, communication, and documentation skills**, preparing us to work confidently on professional full-stack development projects in the future.

10. Conclusion

The PharmaTrack Pro – Intelligent Pharmacy Inventory Management System demonstrates how digital technology can improve pharmacy operations by making inventory management more efficient, accurate, and secure.

By integrating a Spring Boot backend, PostgreSQL database, and React frontend, the system provides a complete full-stack solution for managing medicines, tracking stock, and recording pharmacy transactions in real time.

The project successfully addresses major issues found in traditional pharmacy management systems such as medication errors, expired stock losses, and lack of real-time visibility. Features like FEFO-based stock rotation, role-based authentication, inventory alerts, and analytics dashboards ensure efficient pharmacy operations and better decision-making.

Developing this system provided valuable hands-on experience in backend development, database design, frontend integration, and system testing. Overall, PharmaTrack Pro demonstrates a scalable and practical solution that can support modern pharmacy management and improve operational reliability.

11. Acknowledgements

We would like to express our sincere gratitude to our **project mentor and faculty members** for their continuous guidance, encouragement, and valuable feedback during the development of the **PharmaTrack Pro system**. Their support and technical insights played a significant role in improving the quality of this project.
We also thank our **institution** for providing the necessary resources, tools, and learning environment that enabled us to successfully complete this project.
Finally, we appreciate the **dedication, teamwork, and collaborative efforts of all team members**, whose consistent contributions helped us successfully design, develop, and present the PharmaTrack Pro system.